

Experiment 3 - Write a Python program to solve the Water Jug Problem.

Program Code

```
from collections import deque

def water_jug(capacity_a, capacity_b, target):
    queue = deque()
    visited = set()

    queue.append((0, 0, []))

    while queue:
        a, b, path = queue.popleft()

        if (a, b) in visited:
            continue

        visited.add((a, b))
        path = path + [(a, b)]

        if a == target or b == target:
            return path

    states = [
        (capacity_a, b),
        (a, capacity_b),
        (0, b),
        (a, 0)
```

```
]
```

```
transfer = min(a, capacity_b - b)
states.append((a - transfer, b + transfer))
```

```
transfer = min(b, capacity_a - a)
states.append((a + transfer, b - transfer))
```

```
for state in states:
    if state not in visited:
        queue.append((state[0], state[1], path))
```

```
return None
```

```
capacity_a = 4
capacity_b = 3
target = 2
```

```
solution = water_jug(capacity_a, capacity_b, target)
```

```
if solution:
    print("Solution found!")
    print()
```

```
for step, state in enumerate(solution):
    print("Step", step, ":", state[0], "litres,", state[1], "litres")
else:
    print("No solution found.")
```

Program Output

```
PS D:\AI LAB> & C:\Users\Student\AppData\Local\Programs\Python\Python313\python.exe "d:/AI LAB/experiment3_water_jug.py"
Solution found!

Step 0 : 0 litres, 0 litres
Step 1 : 0 litres, 3 litres
Step 2 : 3 litres, 0 litres
Step 3 : 3 litres, 3 litres
Step 4 : 4 litres, 2 litres
PS D:\AI LAB> 
```